Assignment No 6

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Title – Use DFS for problem solving. Examples can be water jug problem , Missionaries and Cannibals,maze solving, blocks world etc.

Program:-

```
def dfs_maze(grid, start, goal):
    rows = len(grid)
    cols = len(grid[0])

    visited = set()
    path = []

    def dfs(r, c):
        # Check boundaries
        if not (0 <= r < rows and 0 <= c < cols):
            return False

        # Check wall or visited cell
        if grid[r][c] == 1 or (r, c) in visited:
            return False

        visited.add((r, c))
```

```
    path.append((r, c))

    # Goal reached
    if (r, c) == goal:
        return True

    # Down, Right, Up, Left
    directions = [
        (1, 0),
        (0, 1),
        (-1, 0),
        (0, -1)
    ]

    for dr, dc in directions:
        if dfs(r + dr, c + dc):
            return True

    # Backtracking
    path.pop()
    return False

return path if dfs(*start) else None
```

```
# 0 = Open path
# 1 = Wall

maze = [
    [0, 1, 0, 0, 0],
    [0, 1, 0, 1, 0],
    [0, 0, 0, 1, 0],
    [1, 1, 0, 0, 0],
    [0, 0, 0, 1, 0]
]

start = (0, 0)
goal = (4, 2)

path = dfs_maze(maze, start, goal)

print("DFS path:", path)
print("Path found:", path is not None)
```

Output:-

DFS path: [(0, 0), (1, 0), (2, 0), (2, 1), (2, 2), (3, 2), (4, 2)]

Path found: True

Process finished with exit code 0